



UNIVERSITY OF
ARKANSAS

Data Science

Applying a Multiple Layer Perceptron (MLP)

Jonatan Alcala, Kennedy Maloney (Group 18)

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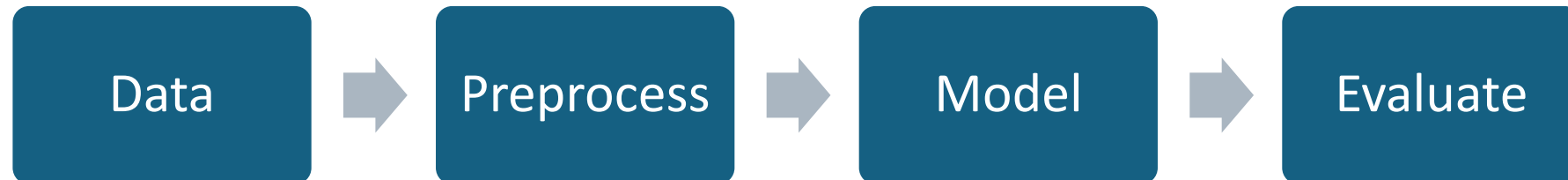
Project Overview

Goal: predict whether a person earns $>50k$ or $\leq 50k$ using demographic data.

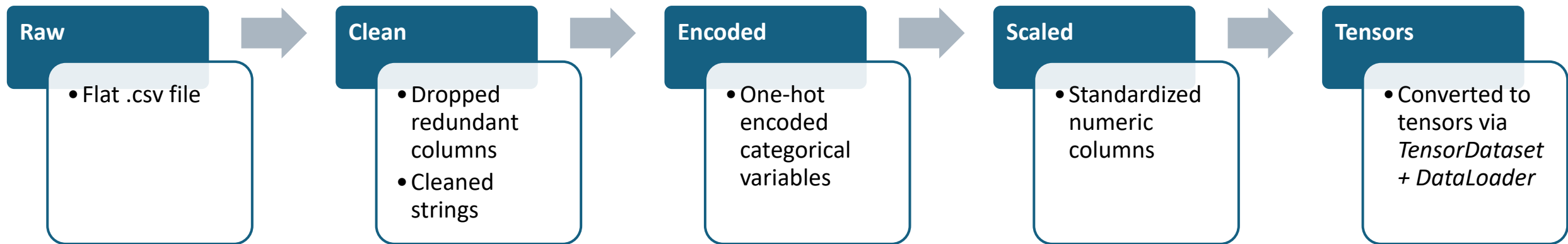
Dataset: UCI Adult Income — mix of categorical & numeric features

Framework: PyTorch MLP

Focus: model design, training behavior, overfitting control, and tuning.



Data Preparation



Model Development

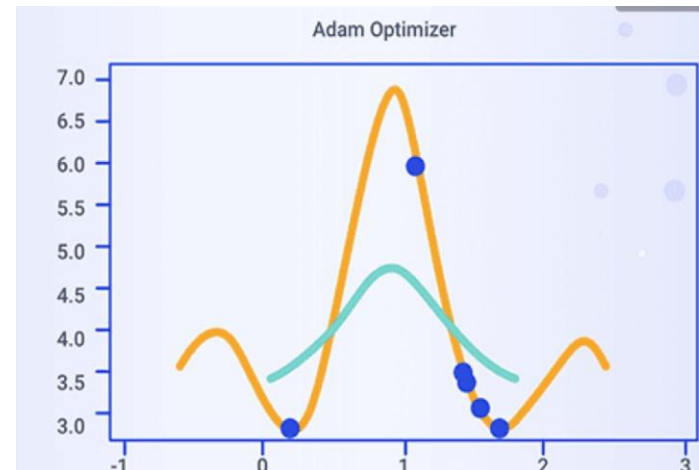
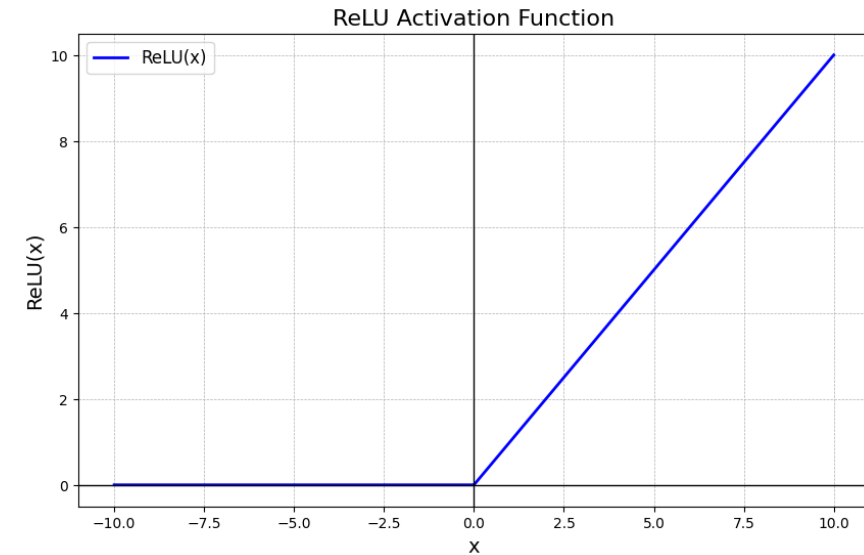
Defined MLP Architecture

- Input: 91 features
- Hidden layers: 128 → 64
- Activation: ReLU
- Dropout: 0.3 to reduce overfitting
- Output: 2 logits (binary classification)

Loss: CrossEntropyLoss

Optimizer: Adam (lr=0.001)

Epochs: 20



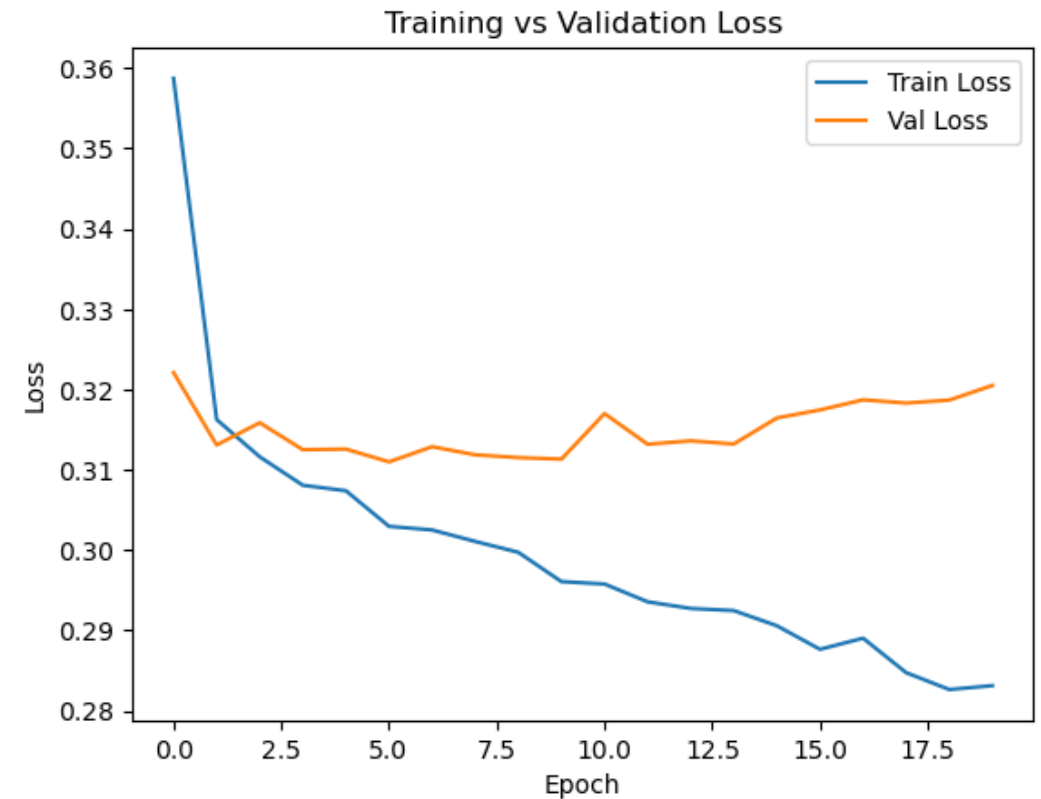
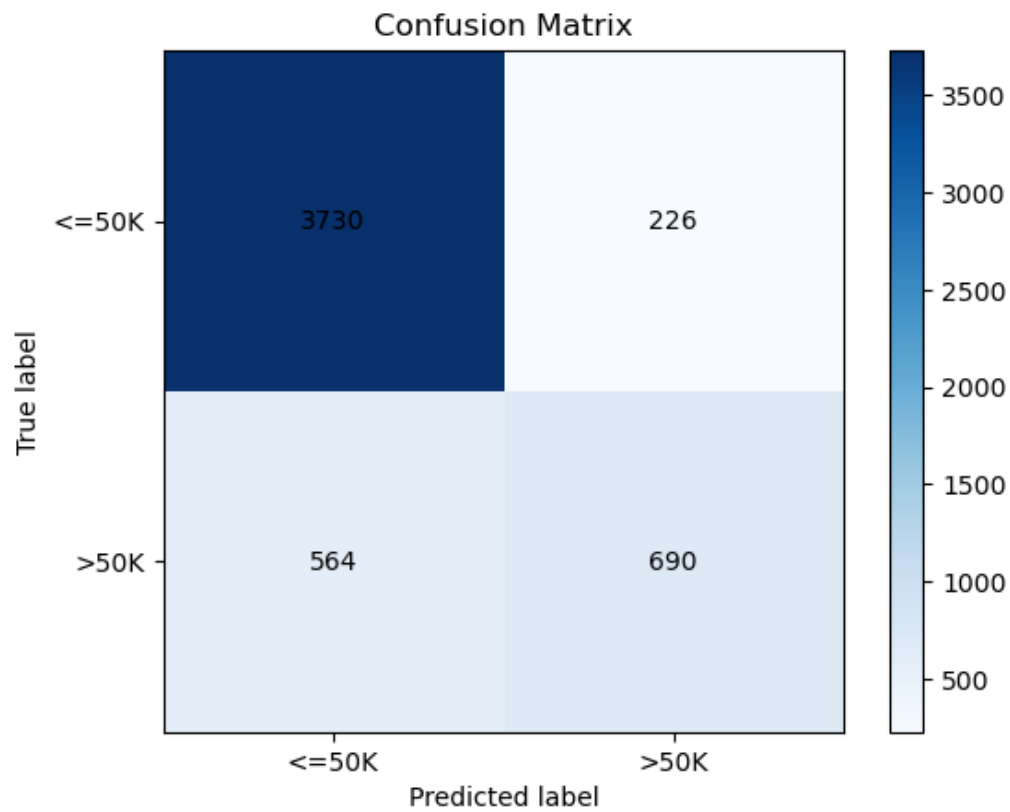
$$H(p, q) = - \sum_{x \in \text{classes}} p(x) \log q(x)$$

True probability distribution (one-shot) $\nearrow$ $p(x)$

Your model's predicted probability distribution $\nearrow$ $q(x)$

Model Development

Stable **~85%** validation accuracy | Overfitting signs appeared after ~epoch 8-10 | Mild divergence after ~10 epochs.



Evaluation & Tunning

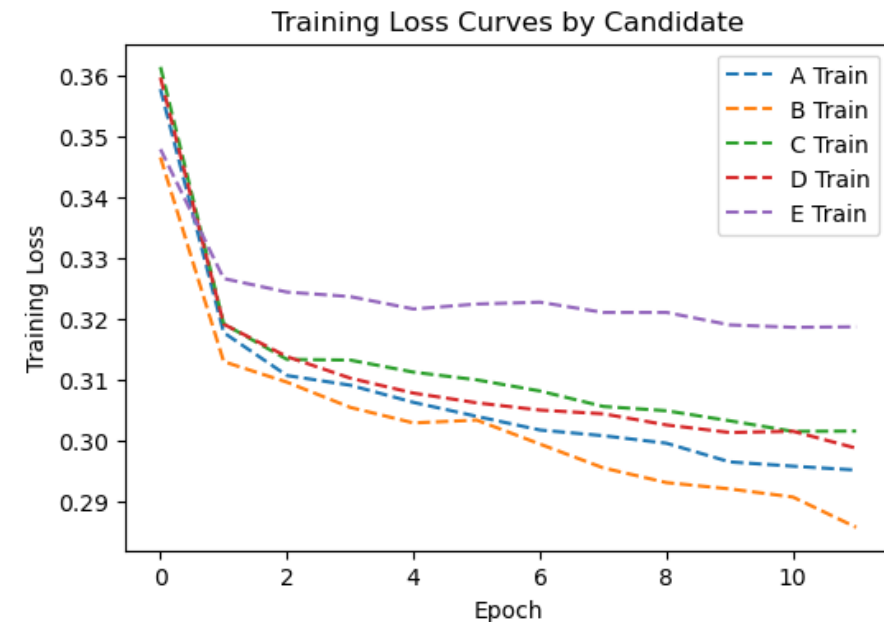
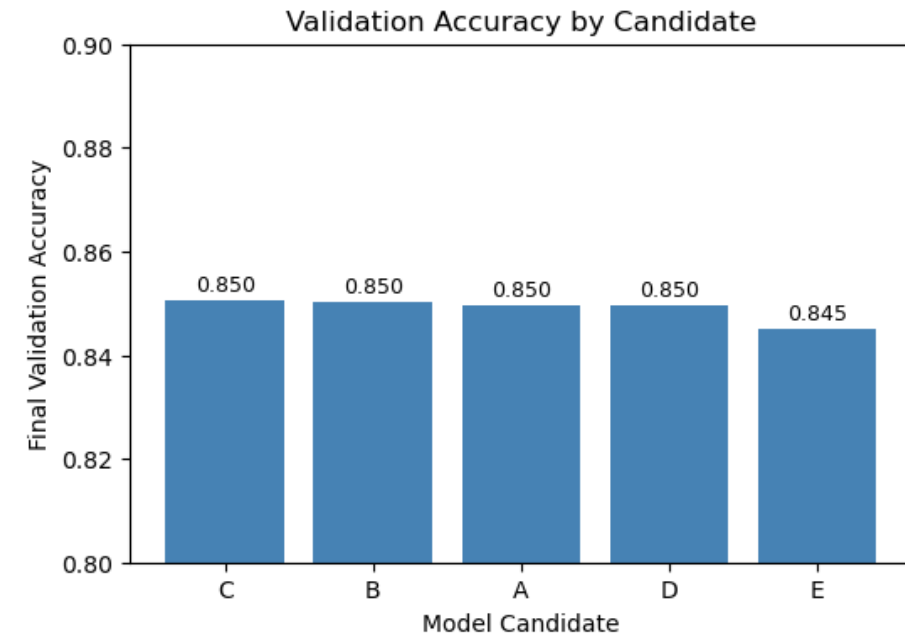
Hyperparameter tuning tested five candidates (A–E) varying:

- Hidden layer sizes
- Dropout rates
- Learning rates
- Activation functions (ReLU, LeakyReLU, Tanh)

Best Candidate*:

- Candidate **B** (256–128, ReLU, dropout = 0.3, lr = 0.001)

*All candidates converged smoothly, showing model stability.



Results & Takeaways

Model Refinement

- Tuned candidate B further & compared to initial baseline model.
- Low-Dropout Model (Dropout = 0.25)
- Class-Weighted Model

Final Model

- **Candidate B** w/ standard dropout (**0.30**) & **no class weighting**.
- Offers best overall trade-off between accuracy (**~0.85**) and precision, ensuring stable, interpretable, and generalizable performance across income categories.

Predict on Validation Inputs

- Used final model to predict the income of a separate validation dataset.

Reflection: Reasoning Behind Model Architectures

Why we choose the specific architecture (e.g., number of layers, activation functions) for each model.

Number of Nuerons: allows the model to understand the broad patterns in the data while remaining sensitive to local patterns

Activation Function: added nonlinearity, performed the best when compared to a Leaky ReLU and Tanh activation function.

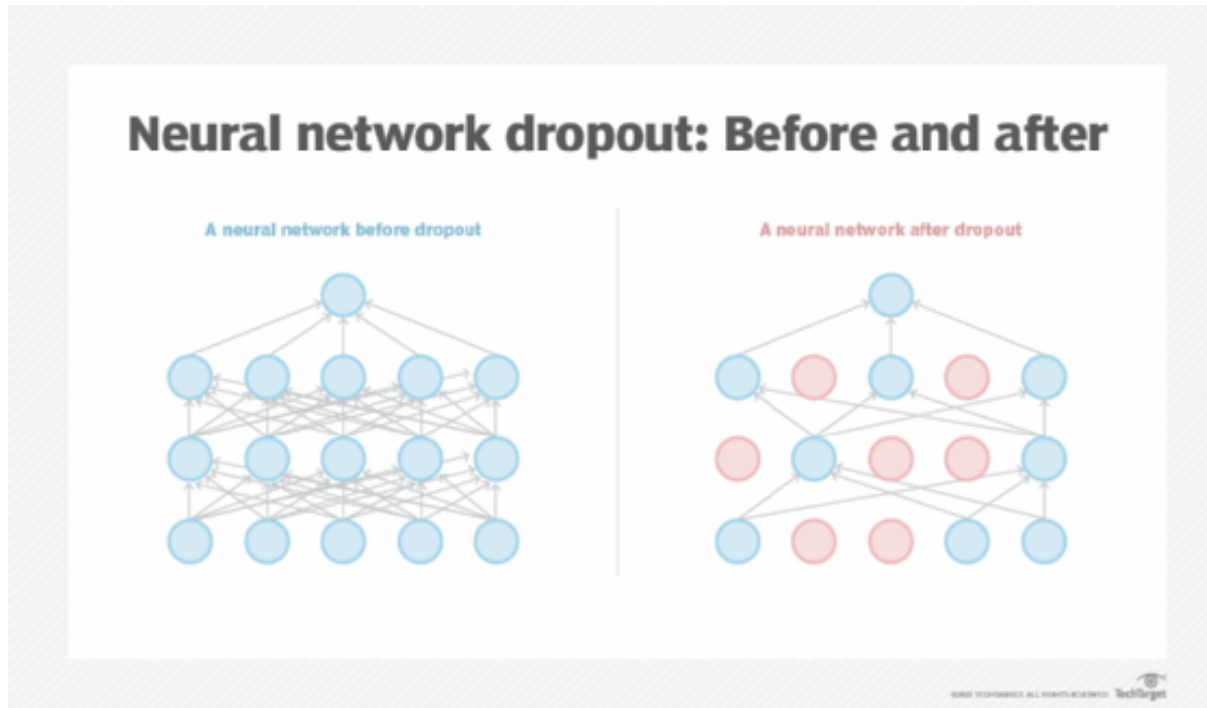
Dropout Level: 30% of the neurons in that layer will be randomly set to zero during each training step to prevent overfitting. Typical range for dropout level is 0.20-0.50.

Binary Output Layer: Works well with Cross Entropy Loss and doesn't overcomplicate the scenario

Reflection: Overfitting

How we monitored and mitigated overfitting in our models.

Using Dropout Layers



Honorable Mentions:

- Using a ReLU activation function
- Using a controlled training duration of 20 Epochs
- Using the Adam Optimizer

Reflection: Ethical Concerns

Ethical concerns might arise from deploying models trained on these datasets.

Historical Bias in the Dataset: Adult Income data reflects past inequalities in occupation, income, gender, and race.

Risk of Discrimination in Predictions: The MLP may learn patterns that disadvantage certain demographic groups.

Proxy Variables for Sensitive Attributes: Even if race or gender are removed, the MLP can infer them through related features (e.g., job type, marital status).

Limited Model Explainability: MLPs are not easily interpretable, making it difficult to justify or audit decisions.

Real world harm

Reinforcing stereotypes and systemic disadvantages

Privacy concerns

=

Reflection: Activation Functions

Why activation functions are necessary in neural networks.

Activation functions are how we **introduce nonlinearity** and **capture complex relationships** in a model.

In the real world, **data is messy** and (usually) **not perfectly linear**.

That's why ReLu, Sigmoid, Hyperbolic Tangent, and many more activation functions are essential. **They make the neural networks more feasible for real world data.**

For this project, without nonlinear activation functions, the network would **reduce to a linear model and would be unable to learn complex patterns in the data.**

From slide 5:
Our ReLU Activation Function

